

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8.	(a)	(i)		Enthalpy / energy change when 1 mol of substance is burned (1) Completely / in excess oxygen under standard conditions (1)	2			2		
		(ii)		$\text{C}_2\text{H}_6 + 3\frac{1}{2}\text{O}_2 \rightarrow 2\text{CO}_2 + 3\text{H}_2\text{O}$ (1) Bonds broken $1(\text{C}-\text{C}) + 3\frac{1}{2}(\text{O}=\text{O}) + 6(\text{C}-\text{H}) = 2080.5 + 6(\text{C}-\text{H})$ (1) Bonds formed $4(\text{C}=\text{O}) + 6(\text{O}-\text{H}) = 5974$ (1) $2080.5 + 6(\text{C}-\text{H}) - 5974 = -1561$ $(\text{C}-\text{H}) = 389 \text{ kJmol}^{-1}$ (1) award (3) for cao ecf possible		4		4	1 1 1	
		(iii)		Average used since each individual bond will be in a different environment and therefore have a different strength (1)			1	1		